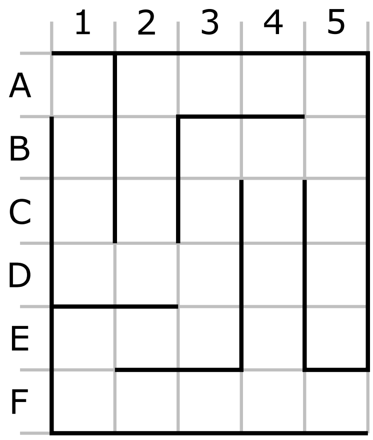
DM879 Artificial Intelligence

Reexam — August 27th, 2021 — Duration: 3 hours

draft solution — answers in blue — comments in green

1 The labyrinth, part I



The figure on the right depicts a labyrinth. The black lines are the walls, and for convenience a light gray grid was added in order to make it easy to refer to cells. The goal of this problem is to find a path from `A1` to `F5`.

One possibility is for example to take the route `A1-B1-C1-D1-D2-C2-B2-A2-A3-A4-A5-B5-B4-C4-D4-E4-F4-F5`, which we can abbreviate to `A1-D1-D2-A2-A5-C5-C4-F4-F5` (by indicating only the places where we change direction), but of course other, shorter, paths are available.

(a) Classify this problem according to the following dimensions: fully/partially observable; deterministic/stochastic; discrete/continuous.

This problem is fully observable, deterministic and discrete.

(b) Formulate this problem precisely as a search problem: describe the state space, the initial state, the actions, the transition model, the goal test, and the cost function.

State space: The positions in the labyrinth (from `A1` to `F5`).

Initial state: `A1`

Actions: Move `left`, `down`, `right` and `up`, as long as there is no wall from that position.

Transition model: Move one step in the corresponding direction.

Goal test: `F5`

Cost function: 1 for each move.

Alternative: move in the same direction as long as there are no alternatives; the cost is then the distance travelled.

(c) Assume that the possible moves are generated in the order `up`, `left`, `down`, `right`, and that duplicate states are skipped.

i. Write the solution found by depth-first search.
`A1-D1-D2-A2-A5-B5-B3-E3-E1-F1-F5`

ii. Write the solution found by breadth-first search.
`A1-D1-D2-D3-B3-B4-F4-F5`

iii. Write the solution found by bidirectional search using uniform-cost search.
`A1-D1-D2-D3-E3-E1-F1-F4-F5`

iv. Propose a (simple) admissible heuristic for this problem.
Manhattan distance to `F5`

v. Write the solution found by best-first search using your heuristic.
`A1-D1-D3-E1-F1-F5`

(d) Formulate this problem as a planning problem in PDDL style: introduce the relevant predicates for describing the state, and indicate the preconditions and effects of action up (moving one step in the "upwards" direction).

We use arithmetic and two additional predicates: At(X, Y), indicating that the agent is at cell in position (X, Y) of the grid, and Wall(X, Y, X', Y'), indicating that there is a wall between cells (X, Y) and (X', Y').

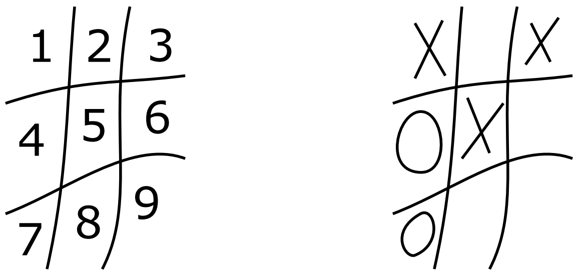
Action up has the preconditions $\{\text{At}(X, Y), Y > 1, \neg\text{Wall}(X, Y, X, Y-1)\}$ and the effects $\{\neg\text{At}(X, Y), \text{At}(X, Y-1)\}$.

2 The student's revenge

After a series of defeats playing tic-tac-toe against the teachers, one diligent student decided to implement an agent for playing the game. As a first step, the student focused on representing the game using first-order logic. The representation is as follows.

- The squares are represented by numbers (1–9), as in the diagram on Figure (a) below.
- There is a (unary) predicate Empty such that Empty(N) holds iff square N is empty.
- There is a (binary) predicate Marked such that Marked(P, N) holds iff square N is marked with the symbol P.
- There are auxiliary (unary) predicates Turn and Won such that Turn(P) holds iff the next player is the player playing symbol P, and Won(P) holds iff the player playing symbol P has won.

In all the examples above, the domain for variable P is the set $\{\text{X}, \text{O}\}$.
Important. This exercise assumes the international (a.k.a. "normal") way of playing tic-tac-toe, where players alternate playing 'X' and 'O' on an empty square until one of them reaches three-in-a-row (win) or all squares are filled (draw).



(a) (b)

(a) Give the complete set of facts that describe the initial state of a new game.

$\{\text{Empty}(1), \text{Empty}(2), \text{Empty}(3), \text{Empty}(4), \text{Empty}(5), \text{Empty}(6), \text{Empty}(7), \text{Empty}(8), \text{Empty}(9), \text{Turn}(\text{X})\}$

(b) Give the complete set of facts that describe the state in Figure (b) above.

$\{\text{Marked}(\text{X}, 1), \text{Empty}(2), \text{Marked}(\text{X}, 3), \text{Marked}(\text{O}, 4), \text{Marked}(\text{X}, 5), \text{Empty}(6), \text{Marked}(\text{O}, 7), \text{Empty}(8), \text{Empty}(9), \text{Turn}(\text{O})\}$

(c) Write two formulas stating that (i) a player wins by marking all cells in the first row and (ii) a player wins by marking all cells in the main diagonal.

(i) $\forall P.(\mathsf{Marked}(P, 1) \wedge \mathsf{Marked}(P, 2) \wedge \mathsf{Marked}(P, 3) \rightarrow \mathsf{Win}(P))$

(ii) $\forall P.(\mathsf{Marked}(P, 1) \wedge \mathsf{Marked}(P, 5) \wedge \mathsf{Marked}(P, 9) \rightarrow \mathsf{Win}(P))$

(d) All possible actions in this game can be written as $\mathsf{Play}(P, N)$, corresponding to player with symbol P playing in square N. Write the PDDL specification of this action in the language used in this exercise.

Precondition: $\{\mathsf{Empty}(N), \mathsf{Turn}(P)\}$; effect $\{\neg\mathsf{Empty}(N), \mathsf{Marked}(P, N), \neg\mathsf{Turn}(P), \mathsf{Turn}(P')\}$ where P' is the other player.

(e) Write the four possible moves for the next player in the game depicted in Figure (b) above and show that for each of them there is a rational response leading to a win by the player playing X. (Your argument should use the effects of the relevant actions to describe the resulting states, showing that Win(X) holds in all of them.)

The four moves are $\mathsf{Play}(\mathsf{O}, 2)$, $\mathsf{Play}(\mathsf{O}, 6)$, $\mathsf{Play}(\mathsf{O}, 8)$ and $\mathsf{Play}(\mathsf{O}, 9)$. The first move can be followed by $\mathsf{Play}(\mathsf{X}, 9)$, which leads to a state where $\mathsf{Marked}(\mathsf{X}, 1) \wedge \mathsf{Marked}(\mathsf{X}, 5) \wedge \mathsf{Marked}(\mathsf{X}, 9)$ holds, from which it follows that $\mathsf{Win}(\mathsf{X})$. The other moves can be followed by $\mathsf{Play}(\mathsf{X}, 2)$, which leads to a state where $\mathsf{Marked}(\mathsf{X}, 1) \wedge \mathsf{Marked}(\mathsf{X}, 2) \wedge \mathsf{Marked}(\mathsf{X}, 3)$ holds, from which it follows that $\mathsf{Win}(\mathsf{X})$.

3 The labyrinth, part II

Now we consider the scenario where the agent does not know anything about the labyrinth except for its size, and has to find a path from cell `A1` to cell `F5`.

Model this problem as a Reinforcement Learning process specifying:

- the task environment with respect to the dimensions: fully/partial observable, deterministic/stochastic, episodic/sequential, discrete/continuous;
- the elements defining the RL process for this setting;
- an example of solution to the problem (a wordy description is also acceptable provided it is precise enough to avoid ambiguities).

Provide also a qualitative description on how the agent will find a solution to the problem.